DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**CENTRAL VENOUS PRESSURE MONITORING**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-PICU-008 (v3)**		
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1.0 PURPOSE

1.1. To maintain a basis for fluid replacement in seriously ill patient.

1.2. To estimate blood volume deficit

1.3. To determine pressure in the right atrium and central veins.

1.4. To evaluate circulatory failure.

2.0 DEFINITION/ABBREVIATION

2.1. **Central venous pressure (CVP)** - monitoring is continuous measurement of pressure within the superior vena cava and reflects the pressure under which blood returned to the superior vena cava and right atrium. It is more accurate than blood pressure monitoring because changes in circulating volume will be reflected in changes in CVP.

2.2. **Normal CVP**: 5 - 12 cm H2O or 2-6 mmHg

2.3. **CVP elevated** measurements indicates an increased in blood volume as a result in sodium and water retention, excessive IV fluids, alteration in fluid balance or renal failure, heart failure or PA stenosis which limit venous outflow.

2.4. **CVP decreased** measurement indicates a decrease in circulating blood volume and may be due to hypovolemic shock hemorrhage or severe vasodilation with pooling of blood in the extremities that limits venous return and to fluid imbalances, negative pressure breathing which occurs when patient demonstrates retraction.

3.0 RESPONSIBILITY

3.1 **Physician** is responsible for inserting central line via Subclavian, Jugular, femoral vein for accurate Central venous pressure monitoring and order needed intervention base on CVP reading.

3.2 **Head Nurse** is responsible to make sure that all needed equipments are available in the unit and supervise all nurses in the unit is competent in caring patient with CVP. They are responsible to carry out physician's orders competent.

3.3 **Charge Nurse** is responsible to check physician's order if carried and reconfirm for any abnormal findings observed prior informing the physician.

3.4 **Staff Nurses** are responsible to carry out physician's orders, refer for any abnormal findings and knowledgeable in performing CVP monitoring

4.0 POLICY

4.1. Reading CVP must be performed by a qualified PICU staff.

4.2. Standard precaution must be observed.

5.0 PROCEDURE (S)

5.1. *Preparatory phase*

5.1.1. Identify the correct patient using IPSG goal 1: To avoid false identification of the patient.

5.1.2. Hygienic hand washing.

5.1.3. Assemble needed equipment.

5.1.4. Evaluate the patient's prothrombin time, partial thromboplastin time and complete blood count.

5.1.5. Explain the procedure to the patient and ensure informed consent is obtained.

5.2. *Performance phase*

5.2.1. Place the infant in the identified position and confirm the zero point.

5.2.2. Position the zero point at the level of the heart.

5.2.3. Turn the stopcock so that the IV solution in the manometer flows into the patient.

5.2.4. Observe the fall in the height of the column of fluid in manometer. Record the level at which the solution stabilizes or stops moving downward. Record CVP and the position of the patient.

5.2.5. Assess the patient's clinical condition. Frequent changes in measurement will serves a guide to detect whether the heart can handle its fluid load and whether hypo or hypervolemia is present.

5.2.6. Turn the stopcock again

5.3. *Follow up phase*

5.3.1. Document readings and patient response.

6.0 EQUIPMENTS

6.1. None

7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

8.1. MOH Guidelines for Pediatric Care

8.2. DPP-PICU-0009(V2) – Central Venous Pressure Monitoring – PMSH, 15/12/1439

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8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) PICU standards, 3rd version, January 2016

9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer

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